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Title Ai-Orchestrated Idle Management for Multi-Engine Drilling Rigs: Reducing Emissions and Fuel Waste in High-Idle Environments

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Abstract

This paper presents a novel Artificial Intelligence (AI)-driven idle management system for oil drilling rigs equipped with multiple diesel generators. Using a drill rig operating with five Caterpillar 3512 engines—as a reference case, the objective is to demonstrate how a coordinated stop-start logic can dramatically reduce idle time, fuel waste, and emissions. The approach addresses the challenge of high idle shares (up to 80%) commonly observed in rotary drilling operations and aligns with Net Zero goals.

The system integrates advanced machine learning algorithms with rig operational telemetry to manage engine cycles dynamically. Each generator's load profile, runtime history, and thermal state are monitored in real time via a hybrid edge-cloud analytics platform. The algorithms orchestrate staggered shutdowns and synchronized restarts to ensure uninterrupted rig power while minimizing parallel idling. Cold start support systems (such as preheaters) maintain engine readiness. A digital twin of the power system models future demand patterns to enable predictive control. The smart controls continuously optimize engine allocation based on performance, redundancy, and energy efficiency parameters.

Assuming 80% idle time and five CAT 3512 engines (1,800 BHP each), idle fuel consumption may exceed 40 liters/hour per engine. Over 24/7 operations, this totals ~0.65 million liters/year in wasted diesel. The implemented idle management system enabled selective shutdowns, cutting idle runtime by 70%. Annual fuel savings exceeded 0.5 million liters, equating to over 1,300 tons carbon dioxide equivalent CO₂e avoided. Operational continuity was maintained through intelligent prioritization and automated load balancing. The smart controls effectively redistributed load during drilling and tripping cycles, ensuring no loss in torque availability or safety margin. Unlike conventional load-following controllers, the AI orchestrator applies machine learning-based load forecasting and operational state classification. By analyzing real-time SCADA data streams (engine RPM, requested fuel, torque), the system identifies upcoming load peaks and orchestrates staggered engine start-stop sequences. Engine maintenance intervals extended by over 40%, reducing Operational Expenditure (OPEX) and increasing uptime. Real-time visualization of emissions and efficiency Key Performance Indicators (KPIs) improved crew awareness and accountability. Unlike single-engine applications, multi-engine rigs required a highly intelligent coordination logic to avoid power drops during high-demand phases. The system successfully addressed this complexity and demonstrated scalability across other land and offshore rigs.

This work introduces the first real-world application of multi-engine idle orchestration using smart controls in land-based drilling rigs. By combining predictive modeling, intelligent scheduling, and operational analytics, it redefines how distributed power assets can be optimized for sustainability. The system provides an actionable path toward Net Zero compliance, with proven economic and environmental gains—even in legacy rigs operating under continuous high-load conditions.

Introduction

The operation of electric drilling rigs is characterized by a high proportion of idle times. These are mainly due to the non-continuous uniform energy demand in the drilling process itself. During the actual drilling phase, load peaks occur when the drill encounters resistance in the rock. These mechanical counterforces cause sharply increasing power demands on the top drive and the pumps in the short term. The performance peaks are even more pronounced during so-called "tripping" - the retraction or extension of the drill pipes. This process requires high drive power from the Drawworks but is a rare but very demanding operating condition for a short time.

Recent studies in fracturing [4][5][6] and offshore drilling [3] emphasize the potential of AI-enhanced control for distributed power assets. However, while hydraulic fracturing fleets have demonstrated that principle even on hybrid architectures, drilling rigs still predominantly rely on conventional unmanaged diesel generator sets without predictive control. This paper addresses this gap. Figure 1 shows a typical load profile [3]

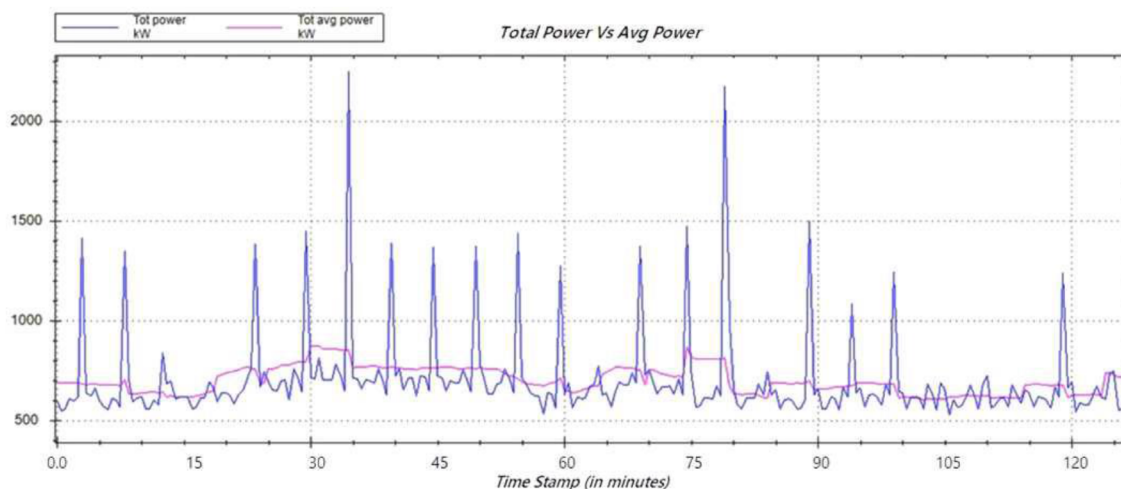


Figure 1—Typical Load Profile of drilling operations [3]

The challenge is that the exact timing of such load peaks is difficult to predict. At the same time, classic diesel generators in the power class of 1 to 1.5 megawatts need many seconds/minutes for a cold start and further time to synchronize and ensure grid stability. Due to this system-related delay, all main generators are in continuous operation throughout operation, even if the actual power requirement is very low at times.

This strategy leads to a significant waste of resources. At idle, a single generator consumes up to 40 liters of diesel per hour and produces emissions without any significant energy consumption. In addition, the maintenance of these engines is mainly based on the number of hours of operation - regardless of whether the engine was running under load or idling. This not only shortens maintenance intervals but also reduces the service life of the components. The consequence is high operating costs, unnecessary CO₂e emissions and poor energy efficiency.

Technical starting point

The technical basis is the analysis of an electrified land drilling rig, as it is typically used in the operation of service companies.

Microgrid

The plant is based on a microgrid system that is operated entirely with diesel generators. In this specific case, it involves five Caterpillar 3512C generators, supplemented by an emergency power generator, such as the CAT C27 type. Each main generator provides around 1,250 kW of electrical power, at a nominal voltage of 600 V and 480 V and a frequency range of 60 Hz. Energy is distributed via two central power distribution buses (one for each voltage level), from which all main consumers are powered.

Drilling Setup

The energy consumers include the consumers known from drilling operations: Top Drive with about 1200 kW, two mud pumps with 400-800 kW each, the Drawworks with about 1000 kW, shakers and rotary equipment with about 100-300 kW as well as auxiliary units for air conditioning, lighting, control and communication (together 50-150 kW). Most of these consumers are controlled by frequency converters (VFD), which allows flexible speed and power control - but does not include feedback to the power supply system about current load requirements.

Setup and test data

The system assessment is based on real operating data from a one-week measurement campaign, which was recorded at the feed-in point of the microgrid. The datasets were divided into three categories:

- **Drilling:** This phase describes the active drilling process with the top drive running and simultaneous operation of the mud pumps. It represents the phase with the highest continuous decrease in energy.
- **Tripping:** This is where tripping takes place, in which the drawworks are operated to retract or extend the drill pipe. This causes short-term but significant load peaks (Figure 3).
- **Redundancy Idle:** These conditions represent classic idle operation, in which one or more generators remain in operation without an active load - often only for redundancy reasons or to hedge the load.

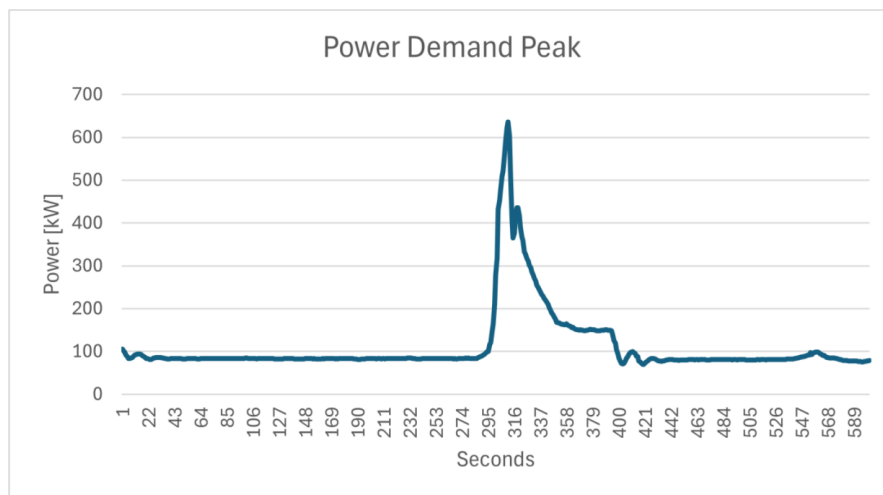


Figure 3—Peak while retracting pipe (own measurement)

Power Factor

The evaluation of the measurement data shows that during the week under investigation, the actual power requirement of the plant rarely exceeded 3.6 MW - although in purely mathematical terms, more than 6 MW of generator capacity was provided in parallel. The average power factor (tPF) was about 0.51, indicating a dominant part-load or no-load drive.

Idle Time

Detailed analysis and classification of the time series showed that about 70-80% of the total operating time took place at idle, with no actual energy consumption. Considering an average idle consumption of 40 l/h per generator, this results in a potentially avoidable diesel consumption of up to 0.78 million liters per year if efficient load management is not implemented. Unnecessary emissions and maintenance costs are correspondingly high, as both service intervals and the service life of components are based on the pure engine running hours - regardless of whether power has been delivered.

In addition, the existing system does not have the technical possibility of temporarily switching off generators during operation and restarting them quickly if necessary. Typical start-up times of classic systems - including preheating, pressure build-up, network synchronization - are beyond 10 seconds and have so far made dynamic load management impossible. There is no coordination between the gensets in terms of redundancy shutdown or power prioritization. There is no central control of load distribution or a mechanism to avoid unnecessary operating hours.

Technical Basis for AI Driven Idle Management

To counter this inefficient mode of operation, clear requirements for a new system architecture must be defined. These include:

- An immediate generator starts in less than 5 seconds, ideally including synchronous grid feed-in.
- Coordinated stop-start operation of multiple generators, adapted to current and forecast loads.
- Synchronization and redundancy management to ensure security of supply even during dynamic operation.
- Avoidance of cold starts, for example through preheater modules and system monitoring.
- Integration of an intelligent control system that automatically determines the best operating point for the entire system based on maps and real-time data.

The technical basis for these requirements is now available. The Engine Standby Controller (ESC) is a proven stop-start technology that puts generators on standby at any time and reactivates them in less than 5 seconds if necessary. In combination with the AI-enabled orchestration platform, a fully networked, intelligent microgrid is created that controls not only individual engines, but the entire system efficiently and proactively.

AI-powered control solution

To substantially reduce the high idle times in electrified drilling rigs, a two-stage control concept was developed based on the combination of local stop-start management and higher-level AI-based load distribution. The central component is the Engine Standby Controller (ESC), which is retrofitted in each individual generator and provides the ability to activate and deactivate autonomously. In conjunction with the cloud-based AI-enabled orchestration platform, this creates an intelligent, networked microgrid that reacts not only to the total energy demand, but also to the dynamic process conditions in real time.

Engine Standby Controller (ESC)

The ESC is a robust, retrofit-capable control unit that has been specially developed for use on diesel-powered generators under extreme environmental conditions. The system makes it possible to put a generator into standby mode fully automatically as soon as there is no active load - and to restart it within seconds if there is a load request. This is made possible by an integrated thermos-management system, active oil circuit management, digital start logic and Li-Ion high power battery support. This reduces classic engine start-up times from 10-20 seconds to less than 5 seconds - including grid synchronization.

Each generator is equipped with a standalone ESC module. This is connected to the higher-level control system and the respective variable frequency drive (VFD) system via standardized interfaces (CAN, Modbus, Digital I/O). In addition, the ESC has its own sensors (e.g. temperature, oil condition, battery voltage) to avoid cold starts and protect the generator hardware. The ESC units operate autonomously locally but continuously exchange status information with the central controller.

AI-enabled orchestration platform - the smart microgrid

The AI-enabled orchestration platform acts as a higher-level, AI-supported control system. It connects all ESC units, energy consumers, measuring points and external data sources (e.g. weather, operational planning) to form a highly available, dynamically controllable smart microgrid. The AI-enabled orchestration platform does not only coordinate the start-stop behavior of all generators but also optimizes the entire energy flow architecture in real time.

By integrating Process data (e.g. drilling depth, pump status, drawworks cycle), Generator status (RPM, load, temperature, preheat status) and consumption patterns on the consumer side, the platform is able to determine the best possible generator combination for each operating condition. The aim is to keep only as many gensets active as are actually needed - while at the same time maintaining an optimal load factor, minimal emissions and high grid stability. The response to load jumps is predictive (preemptive) by recognizing load patterns before they actually occur - e.g. by identifying an upcoming tripping cycle based on previous sequences.

AI-supported map optimization

At the heart of the AI-enabled orchestration platform logic is map-based operating point optimization, which is based on machine learning (ML). A separate operating map is created for each relevant system component:

- Generator maps: fuel consumption (g/kWh), emissions, thermal condition, ageing
- Consumer maps: Energy profiles for Top Drive, Mud Pumps, Drawworks, Shakers, etc.
- System maps: Overall efficiency with different generator combinations and load distributions

These maps are first trained in offline operation with existing measurement data (supervised learning) and then continuously refined in field operation through online learning. The AI system can thus not only detect when certain load patterns occur, but also the state of each individual generator (e.g. temperature level, last maintenance time, number of starts).

Based on this information, the optimal operating point of the entire system is determined. This can shift, for example, if one generator is thermally preloaded or if another is already working more efficiently in the medium load range. In such cases, the AI specifically shifts the load fraction between the generators to maximize overall efficiency while ensuring uniform aging. This enables a dynamic "operating point pooling" approach that optimizes not only for efficiency, but also for system health.

Interfaces and system integration

The overall system is fully compatible with common industrial interfaces. The ESC communicates via digital and analogue inputs/outputs, CANopen or Modbus. The AI-enabled orchestration platform offers

a standardized HMI for operators as well as a connection to higher-level SCADA systems or cloud-based dashboards (e.g. SOPHIA). In addition, data connections to digital drilling planning and process systems is planned.

In the standard configuration, the control is purely autonomous - without active operator intervention. The platform provides the operator with transparent information via visualizations (e.g. generator status, operating hours, efficiency value) and can be manually overridden if necessary.

Results from Data Analysis & Simulation

In order to evaluate the effectiveness of the presented control concept, a comprehensive analysis of the real operating data was carried out. These included high-resolution time series for current, voltage, power, frequency, THD, temperature and other state variables. Based on this data, three clearly definable operating phases could be identified: the drilling phase, the tripping phase and the idle states.

Consideration of the phases

In the "Drilling" phase, the Top Drive and the Mud Pumps were in continuous operation - a classic indicator of active drilling operation. Here, the total energy requirement was up to 3.6 MW, distributed over several generators active at the same time. The "Tripping" phase, on the other hand, was characterized by sporadic, short-term load peaks caused by the raising and lowering of the drill pipe. This phase showed the highest power jump rate per unit of time. In the "Idle" phase, as documented in the data sets, on the other hand, no significant load relationship between generator output and power consumption could be detected over longer periods of time - a clear indicator of inefficient operation.

Reduction of idle time and CO₂ savings

As part of the simulation, the described control system - consisting of ESCs and the AI-enabled orchestration platform - was virtually applied to the same time series. By applying the AI-driven stop-start strategy, the proportion of idle time has been drastically reduced: from an initial 70-80% to only about 15-20%. This equates to a potential saving of approximately 0.78 million liters of diesel per year over the operating time of a typical rig. Using internationally recognized emission factors, this results in an annual CO₂ saving of 2.2 to 2.4 kilotons. Figure 4,5 and 6 shows those efforts.

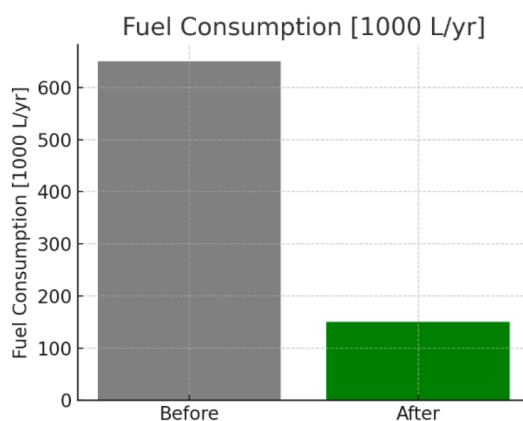


Figure 4—Reduction of Fuel Consumption

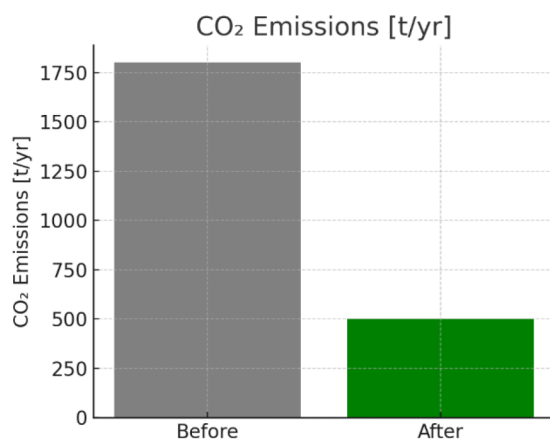


Figure 5—Reduction of CO₂e-Emissions

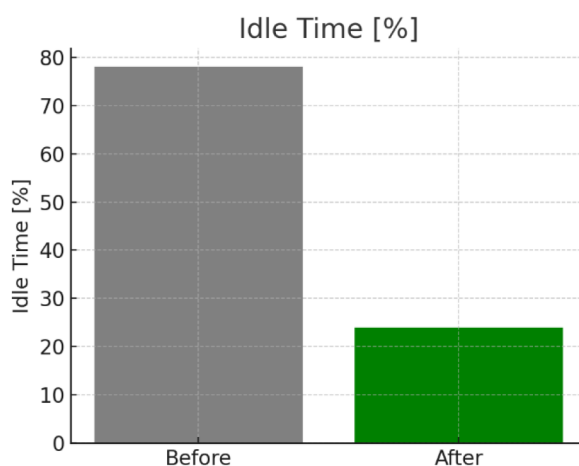


Figure 6—Reduction of Idle Time

Machine Learning capabilities

Another key aspect of the assessment was the quality of the machine learning operating maps. These proved to be highly reproducible and stable across different operating scenarios. The real-time capability was demonstrated by simulating start-stop cycles under dynamic load - including load transfer, synchronization and restart. The system consistently mastered these transitions in less than a second without causing grid instabilities or frequency jumps. There were also positive results in terms of scalability: the control logic could be applied locally on a single device as well as at the system or fleet level.

Payback Period

The simulation results can also be evaluated from an economic point of view. This resulted in an estimated return on investment (ROI) of more than 100 % after 12 months - assuming market diesel prices, average drilling time shares and conservative maintenance costs. The high operational stability in combination with reduced idle times and better utilization of the existing aggregate capacity forms the basis for modern, sustainable energy management at drilling sites worldwide.

Discussion, conclusion and outlook

The solution presented in this paper to reduce idle time in electrified drilling rigs through the use of the Engine Standby Controller (ESC) and the AI-enabled orchestration platform Controls Platform is not limited to onshore drilling rigs but can be transferred to a variety of other applications. Offshore drilling rigs with complex split-bus systems, where generators supply loads via separate grids, similarly benefit from

dynamic stop-start management and real-time control. Even in frac fleets, where a hybrid energy mix of diesel generators and electric pumps is often used, the AI logic presented can make a significant contribution to operational efficiency. Even in mining, for example in electrically powered mining vehicles in open-cast mines, there are comparable requirements: plannable, cyclical load curves, temporarily high energy requirements and considerable downtimes - ideal operating conditions for intelligent control systems.

Main Advantage Driven by AI Technology

A key feature of the overall concept is its continuous ability to learn. The AI models are not only initially trained on historical data but are also constantly optimized during operation. New operating data, changes in the system status or external influences (e.g. climatic conditions) are dynamically integrated so that the system logic adapts independently. The possibility of integrating battery storage systems - both to bridge stop-start cycles and to buffer load peaks - opens additional application possibilities in the field of hybrid energy supply.

The system also offers considerable potential regarding ESG criteria (Environment, Social, Governance). The combination of local monitoring, digital twin and cloud-based evaluation allows CO₂e balancing at machine level. This not only allows sustainability reports to be generated automatically, but also to meet tax and regulatory requirements, e.g. in the context of emissions trading systems or Scope 1 reporting. The fine-grained collection of emission data at component level enables precise, fact-based sustainability control.

Summary

In summary, AI-supported load and start-stop management for diesel generators is not only technically feasible, but also highly attractive from an economic point of view. The solution is retrofit-ready, robust and scalable - making it ideally suited for existing rigs and industrial energy supply infrastructures. The achievable savings in diesel consumption, maintenance costs and emissions make the use profitable in the short term.

The recommendation is therefore clear: the combination of ESC and the AI-enabled orchestration platform should be considered as an overall solution and gradually transferred to operation. AI-based controls will be an integral part of modern power control architectures in the future - comparable to the introduction of VFD systems or digital drilling controls in the past.

Outlook

The outlook shows further innovation potential: Through dynamic re-scheduling based on current drilling plans, the use of semantic models (e.g. LLMs for process pattern recognition) and integration into higher-level energy management systems, control can not only be automated but also optimized. In the long term, entire drilling fields could be operated autonomously - with maximum energy efficiency, minimum CO₂ footprint and full regulatory compliance.

Future work will integrate hybridization layers, leveraging gas gensets, kinetic flywheels or battery buffers realizing downsizing. This would extend the AI orchestrator from idle management toward full hybrid fleet optimization, as already validated in hydraulic fracturing fleets [6].

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